Printout

Tuesday, November 17, 2015 8:58 AM

ELEX2845 MIDTERM 2

Tues, Nov 17th 12:30 PM – 1:20 PM

50 Minutes Total
Aids: One Double-Sided Handwritten Aid Sheet, standard calculator, ruler, pen, pencil, protractor

Name: _____ ID: _____ Set: _____

Multiple Choice – 5 Marks

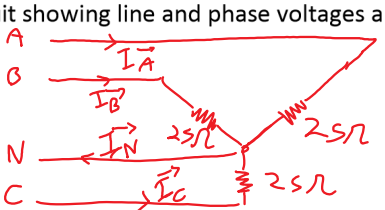
1) To calculate the power in a three-phase circuit, we can use the formula $P = \sqrt{3}|\overrightarrow{V_{LL}}||\overrightarrow{I_L}|cos\phi$. Assuming we don't know whether the load is wye or delta-connected, the angle ϕ is always the phase difference between…
   a. Line-neutral voltage and line-current voltage
   b. Phase A and Phase B line-neutral voltage
   c. Line-line voltage and line current
   d. Phase current and phase voltage
2) Given 3 equal resistors, which connection will result in the **greatest** power dissipation (assume balanced 3P source
   a. Delta
   b. Short-circuit
   c. Wye
   d. Open-circuit
3) In the magnetic circuit analogy, which element is analogous to a resistor in an electrical circuit?
   a. Permeability
   b. Magnetomotive Force (MMF)
   c. Flux
   d. Reluctance
4) A step-up transformer with a 1:2 turns ratio is connected to a 50 V **DC** supply on the primary side. What is the secondary side voltage?
   a. 25 V
   b. 100 V
   c. 0 V
   d. 50 V
5) The connection of a balanced three-phase supply to drive an AC electrical machine does **NOT**…
   a. Establish the direction of rotation
   b. Provide smooth/constant torque
   c. Create a rotating magnetic field
   d. Give a unity power factor

Problem Solving - 15 marks total

Answer the following questions, being sure to show your work where appropriate.

1) A 208V, 60 Hz balanced three-phase source is connected to a load consisting of three 25 Ω resistors in wye-configuration with a neutral path.

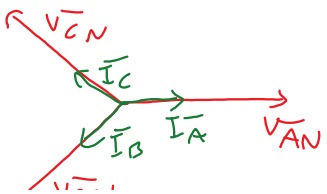
   a) Draw and label the circuit showing line and phase voltages and currents (1 mark)

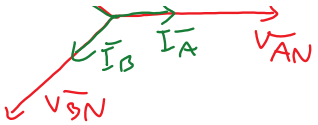


   b) Calculate the line currents in each phase for the balanced load (I_A,I_B,I_C) (1 mark)

$$|\overrightarrow{V_{LN}}| = \frac{208}{\sqrt{3}} = 120\ V \qquad \overrightarrow{I_A} = \frac{120\angle 0^o}{25\Omega} = 4.8\angle 0^o, \overrightarrow{I_B} = 4.8\angle -120^o, \overrightarrow{I_C} = 4.8\angle 120^o$$

   c) Draw a phasor diagram (approximately to scale) showing V_{AN},V_{BN},V_{CN},I_A,I_B and I_C (1 mark)





d) The resistor in phase A is replaced with a 50Ω resistor, determine the new line currents (1 mark)

$$\overrightarrow{I_A} = \frac{120\angle 0^o}{50\Omega} = 2.4\angle 0^o, \overrightarrow{I_B} = 4.8\angle -120^o, \overrightarrow{I_C} = 4.8\angle 120^o$$

e) Calculate the current flowing in the neutral path for the unbalanced load in (d) (1 mark)

$$\overrightarrow{I_N} = \overrightarrow{I_A} + \overrightarrow{I_B} + \overrightarrow{I_C}$$

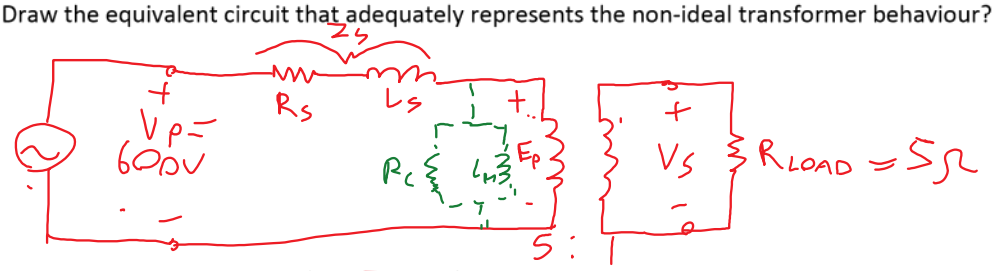
$$I_{Nx} = I_{Ax} + I_{Bx} + I_{Cx}$$
$$= 2.4 + 4.8\cos(-120^o) + 4.8\cos(120^o)$$
$$= 2.4 - 2.4 - 2.4$$
$$= -2.4$$

$$I_{Ny} = I_{Ay} + I_{By} + I_{Cy}$$
$$= 0 + 4.8\sin(-120^o) + 4.8\sin(120^o)$$
$$= 0 - 4.16 + 4.16$$
$$= 0$$

$$\therefore \overrightarrow{I_N} = -2.4\angle 0^o$$
$$= 2.4\angle 180^o$$

2) A single-phase transformer with a 5:1 turns ratio is connected to a 600 V voltage source on the primary side, and a Ω resistor on the secondary. A current of 22 amps of current is measured to flow through the 5 Ω resistor on the secondary side. The real power flowing into the transformer primary is measured at 2500 W. You can neglect Lm and Rc (the magnetizing inductance and core loss resistance) in your answers.

a) Draw the equivalent circuit that adequately represents the non-ideal transformer behaviour? (1 mark)



b) With the resistor connected (switch closed), what is the voltage on the secondary side? Based on this, determi the voltage regulation (assume 22 A on the secondary is full load) (2 marks)

$$V_{FLsec} = I_s R_{LOAD} = (22\ A)(5\Omega) = 110\ V$$
$$V_{NLsec} = 600\left(\frac{1}{5}\right) = 120\ V$$

$$V_{REG\%} = \frac{V_{NLsec} - V_{FLsec}}{V_{FLsec}} \times 100\% = \frac{120 - 110}{110} \times 100\% = 9.09\%$$

c) Based on the measurements taken, determine the series impedance Zs, then Rs and Ls (the series winding loss and leakage reactance). (2 marks)

$$E_p = (110V)\left(\frac{5}{1}\right) = 550\ V$$

$$I_p = (22\ A)\left(\frac{1}{5}\right) = 4.4\ A$$

$$R_s = \frac{P_{LOSS}}{I_P^2} = \frac{P_{IN} - P_{OUT}}{I_P^2} = \frac{2500 - 22^2(5)}{4.4^2} = \frac{80}{4.4^2}$$
$$R_s = 4.13\Omega$$

$$X = \sqrt{Z^2 - R^2} = \sqrt{11.36^2 - 4.13^2} = 10.58\Omega$$

$$I_p = (22\ A)\left(\frac{1}{5}\right) = 4.4\ A$$

$$\therefore Z_s = \frac{(V_p - E_p)}{I_P} = \frac{600 - 550}{4.4} = 11.36\Omega$$

$$X_s = \sqrt{Z_s^2 - R_s^2} = \sqrt{11.36^2 - 4.13^2} = 10.58\Omega$$

$$\therefore L_s = \frac{X_s}{2\pi f} = \frac{10.58}{2\pi 60} = 28\ mH$$

3) A 1500 hp induction motor with 120 poles-per-phase is supplied by a 3-phase, 20 kV, 60 Hz source and draws 50 amps per phase with a power factor of 80% lagging. The slip at full-load is found to be 5%.

a) Calculate the synchronous speed in rpm (1 mark)

$$n_s = \frac{\left(60\frac{s}{m}\right)(60\ Hz)}{p/2} = \frac{3600}{\frac{120}{2}} = \frac{3600}{60} = 60\ RPM$$

b) Calculate the speed of the machine at full-load in rpm (1 mark)

$$slip = \frac{n_s - n}{n_s} \rightarrow n = n_s - n_s(slip) = 60 - (0.05)(60) = 57\ RPM$$

c) Calculate the torque on the shaft at full load in N-m or lb-ft (1 mark)

$$P_{hp} = \frac{T(lb-ft)n}{5252} \rightarrow T(lb-ft) = \frac{5252P_{hp}}{n} = \frac{(5252)(1500)}{57} = (138.2\ x10^3)\ [lb-ft]$$

d) Calculate the efficiency of the motor (1 mark)

$$P_{in} = \sqrt{3}(20\ kV)(50\ A)(0.8) = 1.386\ MW \quad eff = \frac{P_{out}}{P_{in}} = \frac{(1500\ hp)(746)}{1.386MW} = 0.81 \text{ or } 81\%$$

e) Describe how the slip will change if the voltage is **decreased** for the same mechanical load? (1 mark)

Slip would increase (speed would decrease)

Page 2 o